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Git Hands On – 4

Objectives

- Explain how to resolve the conflict during merge.

In this hands-on lab, you will learn how to:

- Implement conflict resolution when multiple users are updating the trunk (or master) in such a way that it results into a conflict with the branch's modification.

Prerequisites

The following are the pre-requisites to complete this hands-on lab:

- Hands-on ID: **"Git-T03-HOL_001"**

Notes*:

Please follow the below steps for creating a free account in GitHub.

Do not use cognizant credentials to login to GitHub.

Estimated time to complete this lab: **30 minutes**.

Please follow the instructions to complete the hands-on. Each instruction expect a command for the Git Bash.

1. Verify if master is in clean state.
2. Create a branch **"GitWork"**. Add a file **"hello.xml"**.
3. Update the content of **"hello.xml"** and observe the status
4. Commit the changes to reflect in the branch
5. Switch to master.
6. Add a file **"hello.xml"** to the master and add some different content than previous.

7. Commit the changes to the master
1. Observe the log by executing **"git log --oneline --graph --decorate --all"**
2. Check the differences with Git diff tool
3. For better visualization, use P4Merge tool to list out all the differences between master and branch
4. Merge the bran to the master
5. Observe the git mark up.
6. Use 3-way merge tool to resolve the conflict
7. Commit the changes to the master, once done with conflict
8. Observe the git status and add backup file to the .gitignore file.
9. Commit the changes to the .gitignore
10. List out all the available branches
11. Delete the branch, which merge to master.
12. Observe the log by executing **"git log --oneline --graph --decorate"**

Solutions :

```
PS D:\Projects\Cognizant\git\gitDemo> git status
On branch master
nothing to commit, working tree clean
PS D:\Projects\Cognizant\git\gitDemo> git checkout -b GitWork
Switched to a new branch 'GitWork'
PS D:\Projects\Cognizant\git\gitDemo> touch hello.xml
touch : The term 'touch' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a
path was included, verify that the path is correct and try again.
At line:1 char:1
+ touch hello.xml
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (touch:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS D:\Projects\Cognizant\git\gitDemo> touch hello.xml
touch : The term 'touch' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a
path was included, verify that the path is correct and try again.
At line:1 char:1
+ touch hello.xml
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (touch:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException
```

```
PS D:\Projects\Cognizant\git\gitDemo> echo "<message>Hello from GitWork</message>" > hello.xml
PS D:\Projects\Cognizant\git\gitDemo> git add hello.xml
PS D:\Projects\Cognizant\git\gitDemo> echo "<message>Updated in GitWork</message>" >> hello.xml
PS D:\Projects\Cognizant\git\gitDemo> git status
On branch GitWork
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   hello.xml

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   hello.xml
```

```
PS D:\Projects\Cognizant\git\gitDemo> git add hello.xml
PS D:\Projects\Cognizant\git\gitDemo> git commit -m "Added hello.xml in GitWork"
[GitWork ef44762] Added hello.xml in GitWork
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 hello.xml
PS D:\Projects\Cognizant\git\gitDemo> git checkout master
Switched to branch 'master'
PS D:\Projects\Cognizant\git\gitDemo> echo "<message>Hello from Master</message>" > hello.xml
PS D:\Projects\Cognizant\git\gitDemo> git add hello.xml
```

```
PS D:\Projects\Cognizant\git\gitDemo> git commit -m "Added hello.xml in master"
[master 55b114c] Added hello.xml in master
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 hello.xml
PS D:\Projects\Cognizant\git\gitDemo> git log --oneline --graph --decorate --all
* 55b114c (HEAD -> master) Added hello.xml in master
| * ef44762 (GitWork) Added hello.xml in GitWork
|/
* 889e356 Added sample file in GitNewBranch
* 2121fb0 Added sample file in GitNewBranch
* c271fbd (GitNewBranch1) initial commit
PS D:\Projects\Cognizant\git\gitDemo> git diff master GitWork
diff --git a/hello.xml b/hello.xml
index 3a8f607..a38c15c 100644
Binary files a/hello.xml and b/hello.xml differ
PS D:\Projects\Cognizant\git\gitDemo> git merge GitWork
warning: Cannot merge binary files: hello.xml (HEAD vs. GitWork)
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.
PS D:\Projects\Cognizant\git\gitDemo> |
```